



SafeNest Guard — AI-Powered IoT Security Monitoring for Bangladeshi Homes

with Instant Bangla Threat Alerts



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Problem Statement

Bangladesh has over 40 million IoT devices in homes, university campuses, and small businesses — and almost none of them are monitored. When someone accesses your home camera from abroad, your router cannot tell if it is you or an attacker on the same ISP. Existing security tools only detect known attack signatures. They have no concept of who the device owner is, how they normally behave, or what trust means in the context of a smart home.

- No affordable IoT security monitoring exists for Bangladeshi homes
- Same-ISP attackers are completely invisible to IP-level filters
- Zero-day attacks bypass every signature-based tool on the market
- Alerts written in English are useless for most Bangladeshi families

SafeNest Guard Stops Big Brother. We see the attacker before your family does.



Then we alert you — in Bangla — with exactly what to do.

Proposed Solution

SafeNest Guard is an AI-powered IoT security system that runs entirely on a BDT 7,000 Raspberry Pi 4 and monitors every device on your home network in real time. It does not just look for known attacks — it learns how your devices normally behave and flags anything that deviates from that pattern. Every 5 seconds, it captures network traffic via tcpdump on the home router, extracts 91 flow features using CICFlowMeter, and runs them through a 5-layer AI pipeline. Four models work in parallel: Random Forest and XGBoost detect known attacks at 99.97% accuracy. Isolation Forest and an Autoencoder catch zero-day threats that have never been seen before. A novel Context-Weighted Fusion layer combines all four detectors, adjusting their weights based on the time of day and the type of device being monitored. The final decision — SAFE, WATCH, THREAT, or REVIEW — is delivered as a clear, actionable Bangla message via Telegram, every time, automatically.

System Architecture

5 Layers on Raspberry Pi 4:

Layer 1 — Capture: tcpdump → CICFlowMeter → 91 flow features every 5 seconds.

Layer 2 — Detection: Random Forest + XGBoost (known attacks) · Isolation Forest + Autoencoder (zero-day).

Layer 3 — Calibration: Isotonic Regression corrects confidence scores using UNSW-NB15.

Layer 4 — CWF + TRAV: Context-Weighted Fusion combines scores by time and device type. TRAV 2.0 scores behavioural trust across 5 dimensions from 7 rolling sessions.

Layer 5 — Alert: SAFE / WATCH / ALERT / REVIEW → Bangla Telegram + live dashboard.

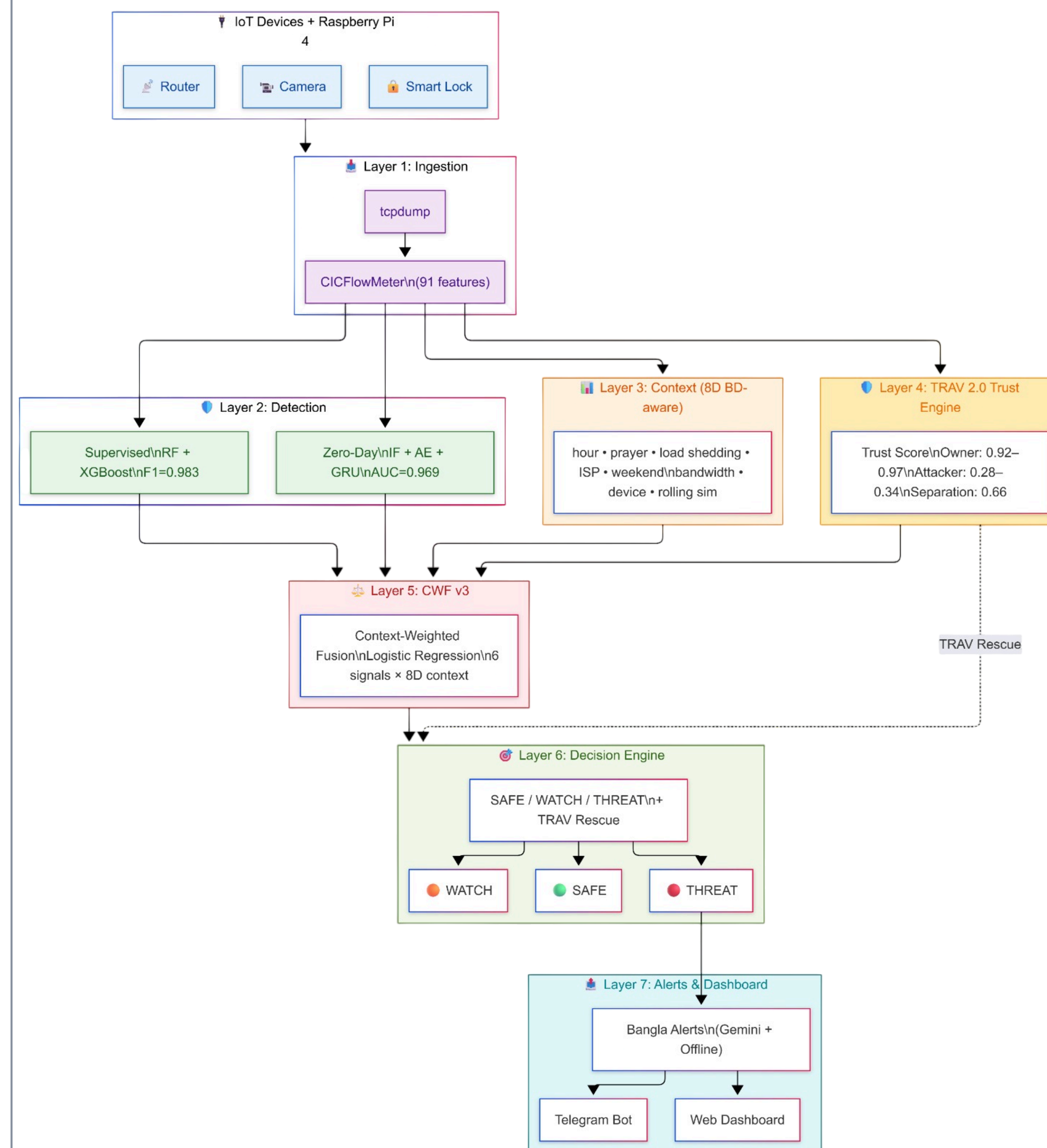


Fig: Safenest System Architecture Diagram

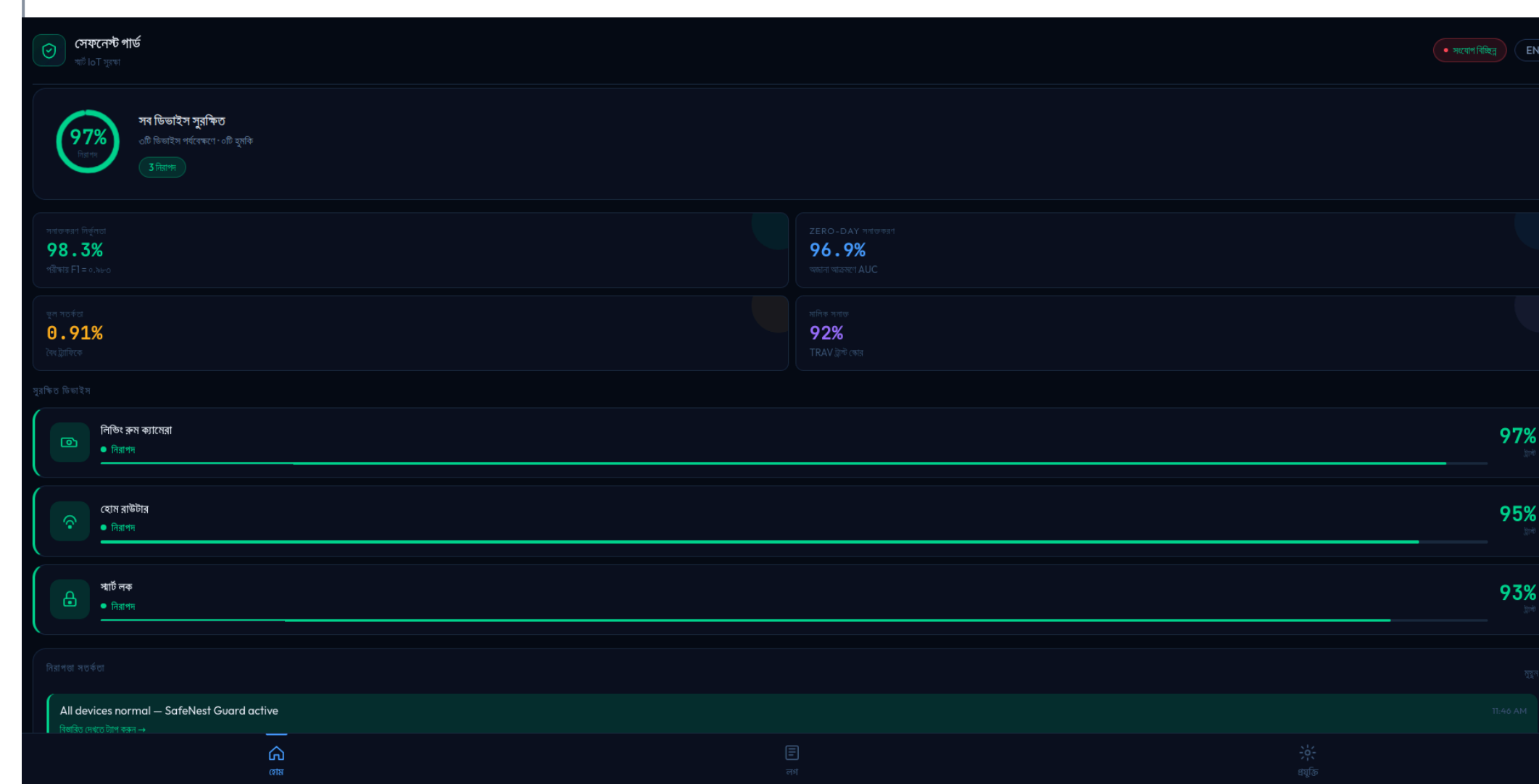


Fig: Monitoring Dashboard

Key Features

- Context-Weighted Fusion (CWF)** An 8-dimensional context vector encoding the hour of day as sin/cos values and the device type modulates the confidence weights of all four detectors through a logistic regression meta-layer. CWF encodes domain knowledge directly: night-time camera access is treated as inherently riskier than the same access at 2 PM. Every decision it produces is fully interpretable and auditable — not a black box.
- TRAV** — Trust & Reputation via Activity Vectors A 5-component behavioural fingerprint maintained across 7 rolling sessions. This is the only IoT security system in Bangladesh that distinguishes the device owner from an attacker by behaviour, not by IP address alone.

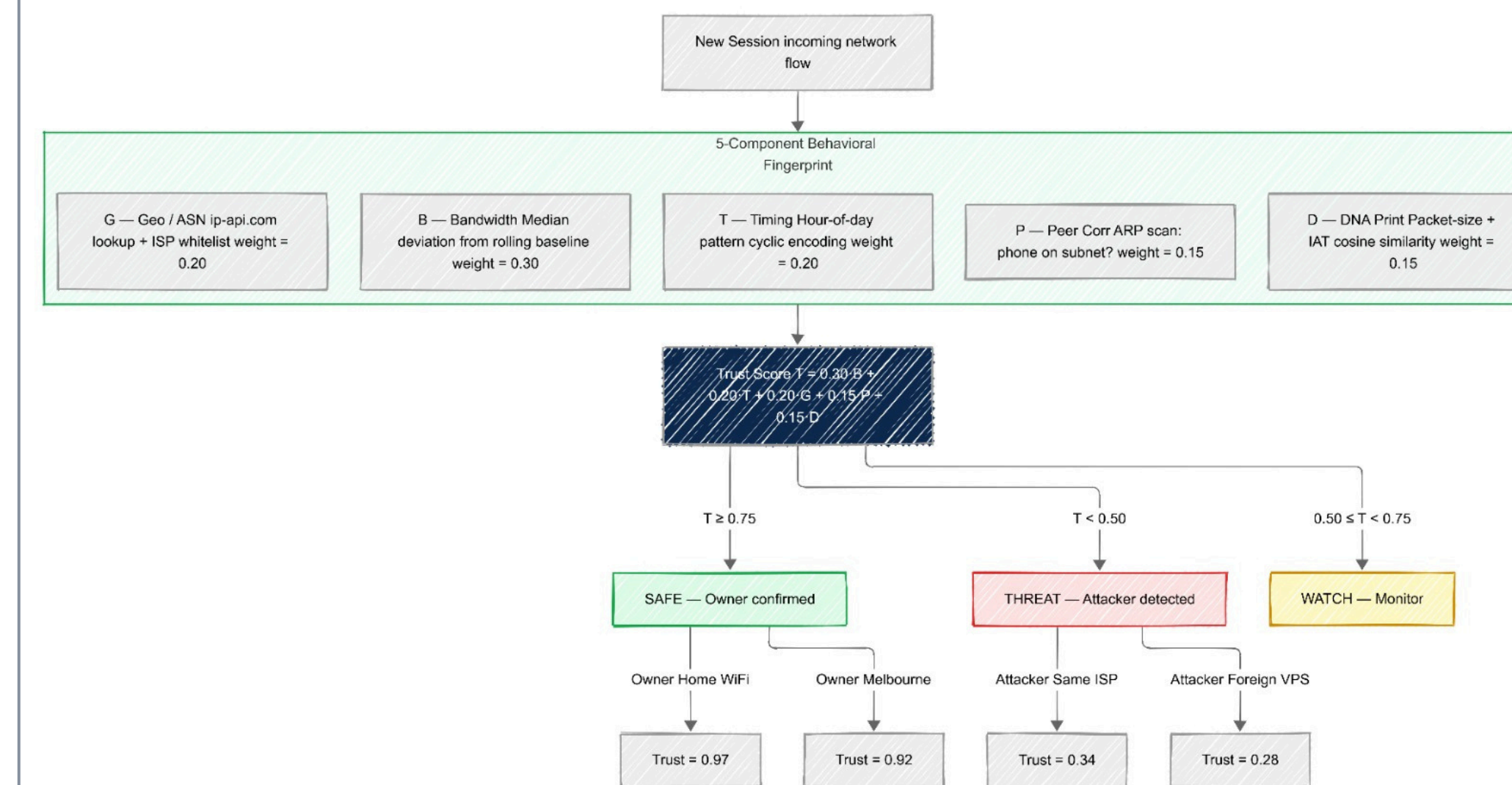


Fig: Trav System Diagram

- Cold-Start New Device Handler** A dedicated NewDeviceHandler class manages devices that appear on the network for the first time. It applies conservative thresholds, bootstraps trust over 3 sessions, and issues a REVIEW decision that asks the owner to confirm the new device via Telegram before full trust is granted. This solves the critical bootstrapping problem that every real-world IoT deployment faces.

Result Overview



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Business Model & Market Readiness

Hardware cost: BDT 7,000 one-time for a Raspberry Pi 4 kit. Affordable for any Bangladeshi household, university department, or small business. Running cost: Near zero. Telegram alerts are free. HiveMQ Cloud free tier covers all MQTT messaging. No cloud servers required. All AI runs locally on the Pi. Three subscription tiers:

- Starter** — Free: Up to 3 devices, Bangla Telegram alerts, 24-hour event history
- Home** — BDT 299/month: Unlimited devices, full TRAV 7-session history, live dashboard
- Business** — BDT 799/month: Multi-site monitoring, API access, priority support, custom alert rules

Gross margin: Above 85% on subscription revenue because infrastructure cost is essentially zero. Market size: Bangladesh has over 40 million IoT devices and the number is growing. Even 0.1% market penetration at the Home tier represents over 40,000 paying subscribers and BDT 12 million in monthly recurring revenue. Competitive advantage: No Bangladeshi competitor exists. International tools like Darktrace or Vectra cost thousands of dollars per month and have no Bangla support, no local ISP knowledge, and no edge deployment option.

Future Enhancement

- Flutter mobile app — block or isolate devices directly from the alert
- Federated learning across SafeNest nodes — better zero-day detection without sharing raw data
- JA3/JA4 TLS fingerprinting — extend TRAV to encrypted traffic
- Full payment gateway integration for subscription tiers
- 10-household Dhaka pilot — real-world longitudinal testing
- Full Bangla language AI assistant for non-technical users

